📄 Project Progress Report (Updated: Day 18)

Project Title: Real-Time Sign Language Recognition System

System Environment: Laptop-based Deployment (Optimized for Portability)

1. Work Completed (Day 1 – Day 18)

- **Architectural Research:** Analyzed CNN vs. MediaPipe-based landmark models.
- **Preprocessing Pipeline:** Finalized techniques for grayscale conversion, normalization, and real-time frame resizing to ensure low-latency performance on consumer hardware.
- **System Design:** Structured the project to handle **static gestures** (Day 10-15) before moving to dynamic sequences.
- **Hand Tracking Selection:** Integrated **MediaPipe Hands** as the primary detection engine due to its lightweight CPU-optimized performance.

2. Implementation Strategy for Laptop Deployment

To ensure your laptop doesn't lag during real-time inference, the system will prioritize **Hand Landmark Data** over **Raw Image Data**. Landmarks reduce the input from thousands of pixels to just **21 coordinate points** per hand.

📊 Sign Language Datasets Summary

Here is the data landscape for your project. I have highlighted the **"Sign Language MNIST"** and **"Landmark CSVs"** as they are the most laptop-friendly due to their small file size and low RAM usage.

Dataset Name	Primary Format	Data Size	No. of Classes	Estimated Memory Usage
Sign Language MNIST	CSV (Grayscale)	~34,000 Images	24 (A-Y, no J/Z)	**Low** (Fits in RAM easily)
ASL Alphabet (Kaggle)	JPEG Images	87,000 Images	29 (A-Z + 3 special)	**High** (Needs Disk Stream)
MediaPipe Landmarked	CSV (Coordinates)	~1,000-5,000 rows	26 (A-Z)	**Minimal** (Fastest for CPU)
ASL_Gestures_36	JPEG Images	2,520 Images	36 (A-Z + 0-9)	**Medium** (Good for testing)

Dataset Name	Primary Format	Data Size	No. of Classes	Estimated Memory Usage
Synthetic ASL	JPEG + Annotations	1,000 Images	26 (A-Z)	**Low-Medium** (Small but high res)

3. Technical Rationale for Laptop Usage

- **Input Resolution:** 28x28 or 64x64 pixels are preferred to keep the CNN parameters low.
- **Data Format:** CSV-based datasets are faster to load on laptops with standard HDDs/SSDs compared to thousands of individual image files.
- **Processing:** By using landmark-based datasets, you bypass the need for a heavy GPU during training.